McMurdo Station, Ross Island

BUILDING INFORMATION SHEET

Vehicle Maintenance Facility

Location ID: B143

Single-story facility used for the maintenance and repair of all vehicles. A high-bay area includes nine repair bays, a welding bay, battery storage room and oil dispensing room. A low-bay area includes offices, a lounge, machine shop, engine repair shop, parts storage, tool storage, restrooms and a boiler room.



Photo 1: *Vehicle Maintenance Facility*

General Structure Information



Automatic External Defibrillator (AED) Locations

This building contains one or more AED devices that may be used in the event of life-threatening cardiac arrhythmia.

- **Mounted on wall between Tool Room and Parts Room.**

Usage	Vehicle Maintenance Facility (VMF)
Occupancy Group	H-4 Repair Garage
Maximum Occupancy	163
Construction Type	Type II, 1-hour, non-combustible; steel-framed structure with 2-1/2" thick foam insulated metal siding and roofing, steel-framed precast concrete slab floor, CMU fire separation wall between high-bay and low-bay areas, steel stud gypsum board interior partitions.
Year Constructed	1987
Dimensions	83' x 249'21' x 40' mechanical room extension

Floor Area

Floor	Area
Floor 1	21,507 sq ft
Total	21,507 sq ft

HVAC System

Heating	One Cleaver-Brooks glycol boiler (?? Btu/hr), one Bryan flex-tube glycol boiler (4,800,000 Btu/hr input) and a glycol heating loop
Air Handling Units	Six central station air handlers
Fuel Tank	8,000 gallon fuel tank

Water and Sewer

Water Service	2" stub off the 4" main
Sewer Service	4" stub off the 4" main
Connections	Sinks: 5 Toilets: 5 Urinals: 0 Showers: 2 Water Cooler: 1 Six lines with twelve instruments
Water Heater	120 gallon water heater, electric

Electrical Service

Service Feeder	Feeder A
Electrical Service	4160V, 3PH, 600A primary disconnect, 4160V/120-208V transformer inside building with 1600A secondary main breaker
Phone and LAN	Six lines with twelve instruments

Fire System

Fire Panel Type	Pyrotronics System 3 control panel, King-Fisher KFRTI-WP-8 radio interface/transmitter panel, heat detectors, pull stations, alarm horn
Heat Detectors	Located throughout building
Suppression Type	The sprinkler system includes a Grinnell 4-inch model F3021 dry pipe valve with accelerator and two mechanical exhausters, two 6,000-gallon tanks with 4,000 gallons each of water at 150 psi tank pressure. The valve releases upon rate of system pressure drop or low air pressure; water is moved by pressure in tanks. Halon System includes Pyrotronics Pyr-A-Lon halon systems in four flammable storage rooms. The heat detector triggers warning, provides a delay, and then releases Halon.

Floorplans

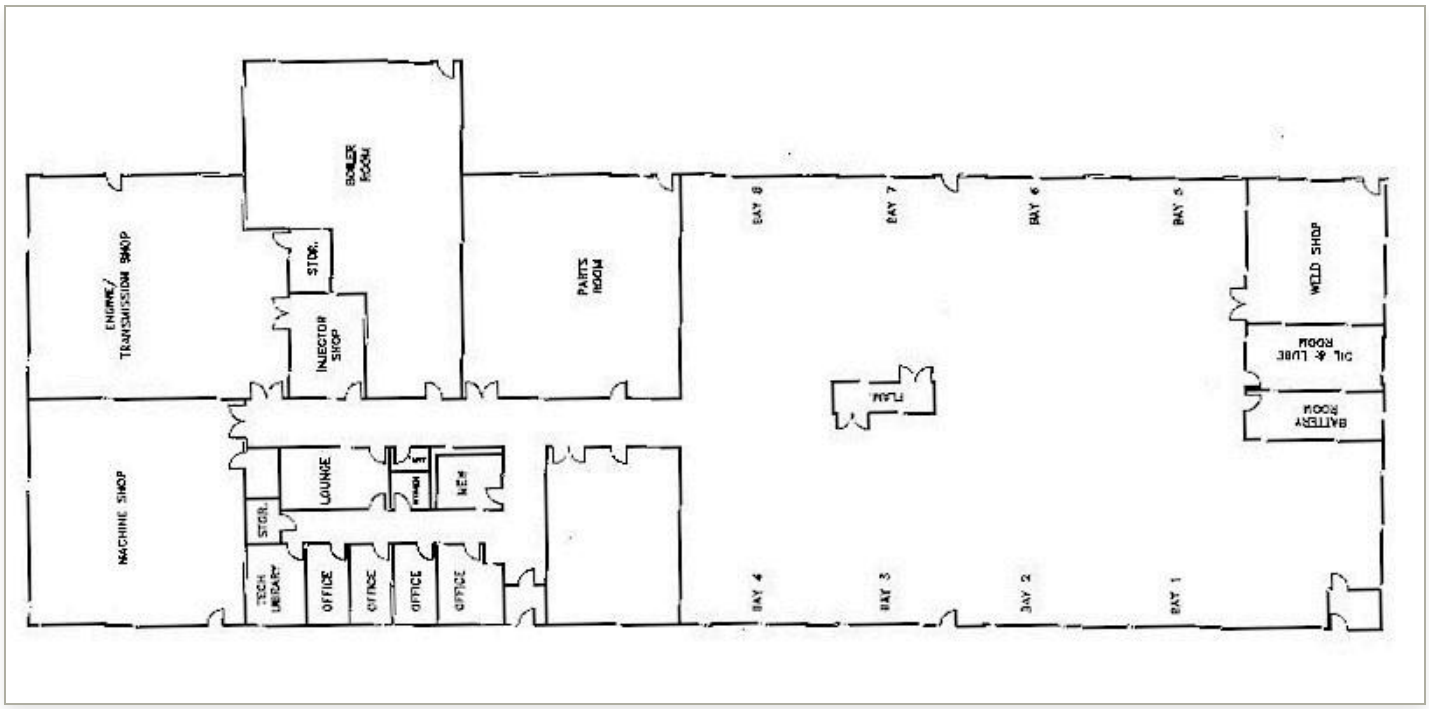


Figure 1: *Vehicle Maintenance Facility*